

Regression Exercise Answers

1. Please enter the *P*-values for marker rs1112 from the analyses in the table below.

	Type of Analysis	<i>P</i> -value
I.	Single marker, case-control, genotypic model	Wald test: het 1/2: 1.40e-05 hom 2/2: 6.24e-06 <u>LRT</u> : 4.077856e-08
	Single marker, case-control, allelic model	<u>Wald</u> : 1.68e-08 <u>LRT</u> : 5.438e-09
II.	Single marker, case-control, adjustment for age	1.68e-08
	Single marker, case-control, adjustment for sex	2.49e-08
	Single marker, case-control, adjustment for sex & age	2.48e-08
	Single marker, case-control, adjustment for marker rs1117	<u>Wald</u> : 0.00144 <u>LRT</u> : 1.193e-03
III.	Single marker, quantitative trait, adjustment for sex	2.29e-14
IV.	Interaction <i>P</i> -value with covariate sex	0.881472
	Interaction <i>P</i> -value with marker rs1117	0.81323

2. Please give the odds ratio (OR) and its 95% confidence interval for marker rs1112 in the unadjusted, genotype-based case-control analysis.

$$OR_{\text{het}(1/2)} = 2.134434 \quad 95\% \text{ CI} = 1.5183466 - 3.0112947$$

$$OR_{\text{hom}(2/2)} = 4.681034 \quad 95\% \text{ CI} = 2.4558674 - 9.4641382$$

3. In the combined analysis of rs1112 and rs1117 (section II., no interaction), the LRT-based *P*-values strongly depended on the order of the markers in the regression model (affection ~ rs1117+rs1112: $p_{\text{rs1117}}=5.547\text{e-}07$ / $p_{\text{rs1112}}=1.193\text{e-}03$; affection ~ rs1112+rs1117: $p_{\text{rs1112}}=5.438\text{e-}09$ / $p_{\text{rs1117}}=0.21$). Do you have an explanation?

Both markers are correlated, i.e. they show allelic association (LD), and represent the *same* phenotypic association signal at the locus (remember that association analysis usually pursues an indirect approach). One of the SNPs, namely rs1112, is more strongly correlated with the causative variant than rs1117. The likelihood-ratio test (LRT) compares the following models: affection~SNP1 vs. affection~SNP1+SNP2. If the marker rs1117 is included in the model first as SNP1, it already contains *some but not all* information on the phenotypic association at the locus. Inclusion of rs1112 as SNP2 still contributes significant additional information. However, if marker rs1112 is included first as SNP1, it already contains *all* information available from the data set on the association of the locus. In this case, inclusion of marker rs1117 as SNP2 cannot contribute any further information and is, thus, tested as being insignificant.